

Welding under pressure

At a training school in the South of France, bubbles form in the water as a student diver begins cutting through metal. The welding process actually takes place within a waterproof, gaseous bubble, which naturally forms around the welder's tool.

MARINE REPAIRS

UNDERWATER WELDING

Welding is the process of joining metals by melting the two separate parts and allowing them to fuse as they cool. It has an important role in construction and sometimes needs to be done underwater—for instance, to make repairs to offshore oil rigs or to fix broken parts of a ship. Underwater welding takes in energy from its surroundings as it happens in what is known as an endothermic process. Safety is crucial, as the metals are melted using electricity.



CONDUCTORS

Water is a conductor of electricity, meaning it allows electricity to pass through it easily. Saltwater is an even better conductor. A saltwater solution can be used to complete a circuit. As underwater welding is often done in saltwater, electric shock is a big risk.

ENDOTHERMIC VS. EXOTHERMIC

Chemical reactions and some processes can be endothermic or exothermic. Oxy-fuel welding (shown here) is different to underwater welding and uses a hot flame to melt metals. This is an example of an endothermic process—one that takes in heat. When the metals solidify, they give out heat to their surroundings. This is an exothermic process.

